

Strata Shares Cooldown Audit Summary

Prepared by Cyfrin

Version 2.0

Lead Auditors

0xStalin

BengalCatBalu

January 23, 2026

Contents

1 About Cyfrin **2**

2 About This Document **2**

3 Disclaimer **2**

4 Copyright and License **2**

5 Audit Overview **3**

6 Audit Scope **3**

7 Risk Classification **4**

8 Protocol Summary **4**

8.1 Share-LockUp Redemption Mechanism 4

8.2 Lock-Up Duration 4

8.3 Withdrawal Requests Finalization 4

9 Attestation **4**

1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

2 About This Document

This document is a public summary of the security audit Cyfrin performed for Strata on Strata Shares Cooldown. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number and final status of findings by severity. It closes with Cyfrin's attestation of the review's scope and final remediation status.

It is deliberately abbreviated: the technical detail of individual findings is not reproduced here. The complete audit report, version 2.0 dated January 23, 2026, was delivered to Strata, who retain it as the full record of the engagement. This summary was issued on October 29, 2026.

3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

4 Copyright and License

© 2026 Cyfrin Inc. All rights reserved. Material from Strata or third parties remains the property of its owners.

This summary may be shared only in complete and unmodified form, so that any copy can be checked against the hash in its attestation. Short quotations with attribution to Cyfrin are permitted, provided they do not misrepresent this summary's scope or results. Except as permitted by applicable law, no part of this summary may otherwise be copied, modified, adapted, sold or sublicensed, or used to create derivative works, without Cyfrin's prior written permission.

Cyfrin and the Cyfrin logo are trademarks of Cyfrin Inc. and may not be used to suggest that Cyfrin has reviewed or endorses anything outside this summary's scope, or to misrepresent its results.

5 Audit Overview

Over the course of 5 days, the Cyfrin team conducted an audit on the Strata Shares Cooldown code provided by Strata. The findings consist of 6 Medium & 3 Low severity issues with the remainder being informational and gas optimizations.

Summary

Project Name	Strata Shares Cooldown
Repository	contracts-tranches
Commit	219880e7a6d4...
Fix Commit	1021020ab866...
Audit Timeline	Jan 9th - Jan 15th, 2026
Methods	Manual Review

Issues Found

Severity	Found	Resolved	Acknowledged
Critical Risk	0	0	0
High Risk	0	0	0
Medium Risk	6	6	0
Low Risk	3	3	0
Informational	10	4	6
Gas Optimizations	2	1	1
Total	21	14	7

6 Audit Scope

The audit scope was limited to changes in the `sip/lockup` branch at commit 219880:

```
contracts/tranches/base/cooldown/SharesCooldown.sol
contracts/tranches/strategies/ethena/sUSDeStrategy.sol
contracts/tranches/Tranche.sol
contracts/tranches/StrataCDO.sol
contracts/tranches/Accounting.sol
contracts/tranches/TwoStepConfigManager.sol
```

7 Risk Classification

	Impact: High	**Impact: Medium**	**Impact: Low**
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

8 Protocol Summary

Strata Money Tranches is a perpetual risk tranching system built around the Ethena ecosystem, specifically around USDe and sUSDe. Strata Tranches is composed of two tranches (Seniors and Juniors), which allows investors to deposit their assets (USDe or sUSDe) into either tranche based on their risk profile.

The newest mechanism is backwards compatible with the existing version of Strata Tranches.

8.1 Share-LockUp Redemption Mechanism

The latest changes consist of a share lockup redemption mechanism that introduces a cooldown period on shares rather than on assets, and charges fees depending on the current `coverage` for unlocked assets.

The exchange rate for the finalization is not fixed to match the exchange rate at the moment of creating the withdrawal request; instead, the exchange rate is determined at the time of finalization.

Tranche shares are not burned at the moment of creating the withdrawal request; they are transferred into a Silo contract and stored until the request is finalized. Shares are burned only when the withdrawal request is finalized.

Withdrawers can opt to cancel a withdrawal request and get back their Tranche shares (minus any fees that have already been paid).

Withdrawers can also opt to instantly finalize their withdrawal requests by paying a fee for skipping the remainder of the cooldown period.

8.2 Lock-Up Duration

The protocol may define up to two coverage thresholds (C_0, C_1) that partition the coverage space into three mutually exclusive ranges, each associated with a predefined lock-up duration.

8.3 Withdrawal Requests Finalization

Expired withdrawal requests can be finalized permissionlessly.

The owner of a withdrawal request that is still under its cooldown period can instantly finalize it by paying a fee.

9 Attestation

Cyfrin attests that it performed a security review of the Strata Shares Cooldown code provided by Strata.

The repositories, commits, review period and findings with their final status are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at https://github.com/Cyfrin/cyfrin-audit-reports. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin's repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2026-01-23-cyfrin-strata-shares-cooldown-v2.0.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin's full audit report with all findings provided to Strata has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name.

```
H=f1c921a1b325e85e25c5c61e4cd3cbdd8d7f4d5ae7d91de7aeb412985e05bc14
F=2026-01-23-cyfrin-strata-shares-cooldown-v2.0-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, `sha256sum -c` can be used in place of `shasum -a 256 -c`. On Windows, run `Get-FileHash <file> -Algorithm SHA256` in PowerShell and compare the result with the hash.